

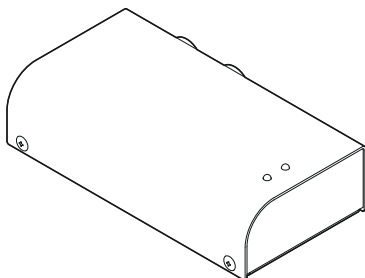
INSTRUCTION MANUAL

Professional tool

delvo

Screw Fastening Monitor

Model: DTM10/DTM45



Indoor Use Only

EN

This instruction manual is written in English.

Instruction manuals in other languages can be downloaded from the URL below.

FR

Ce manuel d'instructions est rédigé en anglais.

Les manuels d'instructions dans d'autres langues peuvent être téléchargés à partir de l'URL ci-dessous.

DE

Bedienungsanleitungen in anderen Sprachen können von der unten genannten URL heruntergeladen werden.

IT

Il presente manuale di istruzioni è redatto in lingua inglese.

I manuali di istruzioni in altre lingue possono essere scaricati dal seguente URL.

ES

Este manual de instrucciones está escrito en inglés.

En la dirección URL indicada abajo se pueden descargar los manuales de instrucciones en otros idiomas.

PT

Este manual de instruções está escrito em inglês.

Pode descarregar os manuais de instruções nos outros idiomas a partir do URL indicado abaixo.

SC

本说明手册的语言是英文。

其他语言的说明手册可从以下 URL 下载。

<http://www.nitto-kohki.co.jp/e/>

[Specifications]

Model	DTM10	DTM45
Communication method	RS-232C	
Mass (kg)	0.26 (0.57)	0.28 (0.62)
Connectable electric screwdriver/ controller models	DLV04C10L-A# DLV10C10L-A# DCC0101X-AZ	DLV45C12#-A# DCC0241X-AZ

#1: X, Y, Z

#2: P, L

■ Refer to p. 4 for details.

■ For operation of the electric screwdriver, refer to the instruction manual that is provided with the electric screwdriver.

- Please read the manual carefully before you attempt to use your tool so that you may use it properly and safely.
- Keep the manual handy - so you can use it whenever necessary.

• Due to continuous product development/improvement, the specifications and configurations in this document are subject to change without prior notice.

Manufactured by.

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(Original Instructions)
TV08150-2 02/2020

Instructions

Thank you very much for your purchase of this NITTO KOHKI product.

Before using your tool, please read this manual carefully so that you may use it properly to get the most out of it.

Please keep this manual handy - so you can use it whenever necessary.

The following safety notations are used throughout the manual to highlight safety precautions for the user and for the tool.



WARNING:

Indicates a potentially hazardous situation which, if not avoided by following the instructions given, could result in death or serious injury.



CAUTION:

Indicates a potentially hazardous situation which, if not avoided by following the instructions given, could result in injury or material damage.

* Please note, however, that failure to observe safety precautions under the "**CAUTION**" category could result in a serious occurrence depending on the situation.

Please observe all safety precautions in the manual.

CAUTION: Important precautions for tool setup, operation and maintenance.

About pictograms



WARNING:

Failure to follow the instructions for handling could cause danger when using the tool.



Using this tool improperly could result in serious injury. Read this instruction manual before using the tool.

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Precautions on Use (Make sure to follow the instructions given)

WARNING

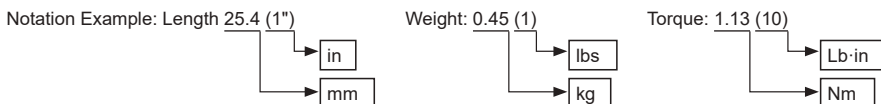
- **Be sure to check the connectable electric screwdriver with this instruction manual.**
If the model of your electric screwdriver is not mentioned in this manual or it is not an electric screwdriver, or it is not our product, do not connect it.
Doing so could cause a fire or accident.
- **Use the tool indoors.**
This tool does not have dust proof or water proof structure. Exposure to dust, sand, or rain while using it outdoors could cause a failure.
- **If the tool is accidentally dropped or hit, check to ensure there is no breakage, cracking or deformation.**
Breakage, cracking or deformation could cause electric shock or injury.
- **When the tool becomes overheated, or you notice anything abnormal during usage, switch it OFF immediately and request that the tool be repaired.**
- **Make sure that there is nothing combustible near the tool before starting work.**
If there is anything combustible near the tool, it could cause a fire or accident.
- **There is an error difference in time measurement of the controller and screw fastening monitor because of a communication lag in DTM10.**
- **The numerical data transmitted by this tool is the converted value of motor current.**
It is not the value of the fastening torque.
- **Control the torque properly.**
Be sure to test more than once based on actual screw fastening conditions (screw, workpiece and other conditions). Perform the test periodically.

Operating Procedure

- (1) Under actual screw fastening conditions, use the screwdriver to fasten the screw.
- (2) Using a torque wrench, etc., make sure the screw is fastened at the target torque. (Check using the loosening torque method or refastening torque method.)
- (3) If the screw is not fastened at the correct torque, adjust the torque of the electric screwdriver and fasten the screw again.
- (4) When the screw is fastened at the correct torque, use a torque checker (torque sensor) to maintain the torque of the electric screwdriver.

About Unit Notation

This instruction manual is written using both SI units and the imperial measurement method (yards, pounds). Numeric values outside the () are the value in SI units, while those inside the () are the imperial measurement value.



About FCC/ISED

- **NOTE:** This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules.
These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.
This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.
- This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:
 - (1) this device may not cause harmful interference, and
 - (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

1. Product Overview

Application

This monitor is a peripheral device that transmits the numerical data of the current control type screwdriver during screw fastening to the PC and Programmable Logic Controller (PLC).

It transmits the following data to the PC or PLC by RS-232C communication.

Operation channel	A unit with which fastening is performed continuously under the same conditions, such as the screw fastening torque, rotational speed, and number of fastening screws ● For details, refer to the instruction manual for the current control type electric screwdriver.
Converted current value	Motor current value at torque up converted to a value between 0 to 4095
Time	Time from the start of the electric screwdriver rotation until torque up (0 to 9990 msec)
Rotation signal (DTM45 only)	Signal from the start of the electric screwdriver rotation until torque up (0 to 60000)

DTM45 outputs time or rotation signal. You can select to output either of the two in "Measurement method" (MEASURE) of the common settings of the controller (DCC0241X-AZ). For controller settings, check the instruction manual of DCC0241X-AZ.

Checking inside the package

When you open the package box, check the contents of the package and also check for any damage caused by incidents during transportation.

If a problem is found, consult with your dealer where you purchased the product.

Screw fastening monitor	DTM10	DTM45
Package content and accessories	Quantity	Quantity
Screw fastening monitor (main unit)	1	1
Connection cord	1	1
Rubber feet	4	4
Instruction Manual (this document, Japanese)	2	2

DTM10



Connection cord LW9075 (2 m long)

DTM45

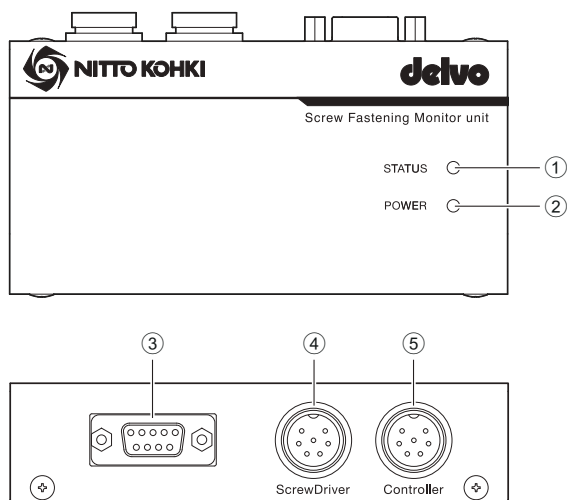


Connection cord DLW9078 (2 m long)



Rubber feet

Part names



Name	Function
① [STATUS] LED	Lights in blue when data is transmitted.
② [POWER] LED	Lights in green when current is applied.
③ RS-232C connector	Used for connection to the PC or sequencer with the RS-232C communication cable.
④ [ScrewDriver] receptacle	Used for connection to the electric screwdriver with the connection cord.
⑤ [Controller] receptacle	Used for connection to the controller with the connection cord.

2. Specifications

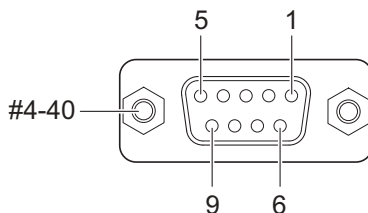
Model		DTM10	DTM45
Communication method	Transmission method	Asynchronous (asynchronous communication)	
	Communication line	Full duplex	
	Transmission speed	38400 bps	19200 bps
	Number of data	8 bits	
	Parity	None	
	Stop bit	1 bit	
	Handshake	None	
Voltage level	HIGH	3 to 15 V	
	LOW	-15 to -3 V	
Mass (weight) (kg)		0.26 (0.57)	0.28 (0.62)
External dimensions		W 124 (4.88) × H 30 (1.18) × D 63 (2.48)	
Compatible models		DLV04C10L-A# ¹ DLV10C10L-A# ¹ DCC0101X-AZ	DLV45C12#-A# ¹ DCC0241X-AZ

#¹: X, Y, Z#²: P, L

RS-232C connector

Specifications	D-SUB 9 pin (female)
Screw	Inch female screw (#4-40)

Pin No.	Name	Input/output direction	Description
2	RxD	Output (Signal output from the monitor)	Transmission data
5	GND		Signal ground



- Other pins are not used

Transmission data

⚠ CAUTION

- SUM value is the sum (including ",",) of each character data from the first byte to the last byte of the transmission data.**
- There is an error difference in time measurement of the controller and screw fastening monitor in DTM10.**

The following data is transmitted to the PC or PLC.

DTM45 outputs time or rotation signal. You can select to output either of the two in "Measurement method" (MEASURE) of the common settings of the controller (DCC0241X-AZ). For controller settings, check the instruction manual of DCC0241X-AZ.

Operation channel	A unit with which fastening is performed continuously under the same conditions, such as the screw fastening torque, rotational speed, and number of fastening screws For details, refer to the instruction manual for the current control type electric screwdriver.
Converted current value	Motor current value at torque up converted to a value between 0 to 4095
Time	Time from the start of the electric screwdriver rotation until torque up (0 to 9990 msec)
Rotation signal (DTM45 only)	Signal from the start of the electric screwdriver rotation until torque up (0 to 60000)

Data format	UTF-8
Data length	11 bytes minimum 19 bytes maximum (variable length)
Delimiter (separator character)	CR + LF (\r\n)
Output timing	When screw fastening is complete

Send command

CH number (3 to 4 bytes)				:	Converted current value (1 to 4 bytes)			
CHp (character data) (p = 1 to 30)					0 to 4095 (character data)			
0x43	0x48	0x30 to 0x39 Tens place ^{*1}	0x30 to 0x39	0x3a	0x30 to 0x39 Thousands place ^{*1}	0x30 to 0x39 Hundreds place ^{*1}	0x30 to 0x39 Tens place ^{*1}	0x30 to 0x39

-	Time (msec) (1 to 4 bytes)					,	SUM value (1 byte)	CR character (1 byte)	LF character (1 byte)
	0 to 9990 (character data)								
0x2d		0x30 to 0x39 Thousands place* ¹	0x30 to 0x39 Hundreds place* ¹	0x30 to 0x39 Tens place* ¹	0x30 to 0x39	0x2c	0x00 to 0xFF	0x0d	0x0a
	Rotation signal (1 to 5 bytes)								
	0 to 60000 (character data)								
	0x30 to 0x36 Ten thousands place* ¹	0x30 to 0x39 Thousands place* ¹	0x30 to 0x39 Hundreds place* ¹	0x30 to 0x39 Tens place* ¹	0x30 to 0x39				

*1 Omitted if the first digit of the CH number, converted current value or rotation signal is 0. The respective places below are omitted in the following examples.

Operation channel: Tens place

Converted current value and time: Thousands place

Rotation signal: Ten thousands place

E.g.) Operation channel: 1, converted current value: 100, screw fastening time: 400 (msec)

byte	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Item	Operation channel			:	Converted current value			-	Time			,	SUM value	CR character	LF character
UTF-8 HEX	0x43	0x48	0x31	0x3a	0x31	0x30	0x30	0x2d	0x34	0x30	0x30	0x2c,	0x74	0x0d	0x0a
DATA	C	H	1	:	1	0	0	-	4	0	0	,	t	\r	\n

● Calculation of SUM value: $0 \times 43 + 0 \times 48 + 0 \times 31... + 0 \times 30 + 0 \times 2c = 0 \times 274$

E.g.) Operation channel: 3, converted current value: 310, rotation signal: 3000

byte	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item	Operation channel			:	Converted current value			-	Rotation signal				,	SUM value	CR character	LF character
UTF-8 HEX	0x43	0x48	0x33	0x3a	0x33	0x31	0x30	0x2d	0x33	0x30	0x30	0x30	0x2c	0xa8	0x0d	0x0a
DATA	C	H	3	:	3	1	0	-	3	0	0	0	,		\r	\n

● Calculation of SUM value: $0 \times 43 + 0 \times 48 + 0 \times 31... + 0 \times 30 + 0 \times 2c = 0 \times 2a8$

● There is no corresponding character for SUM value "0xa8".

3. Connection Diagram

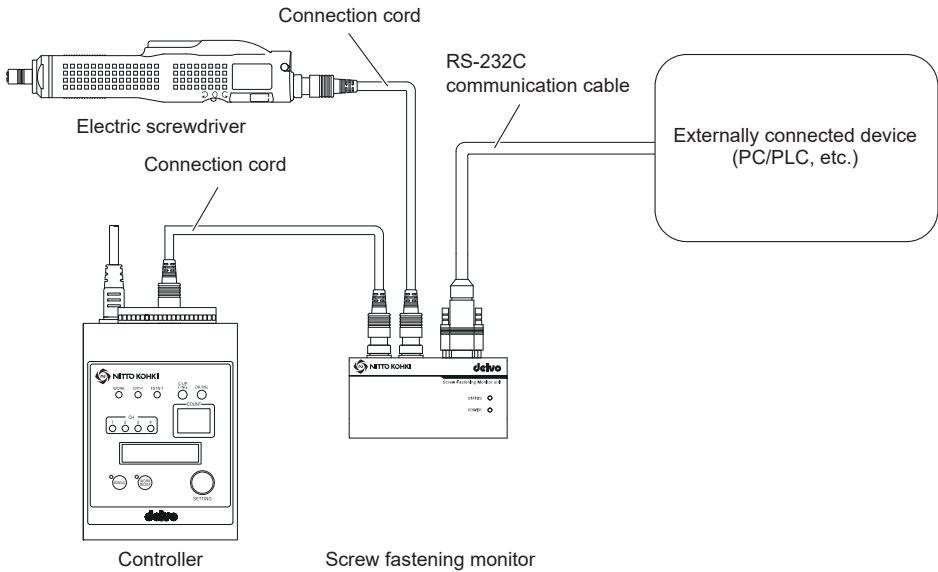
This is explained with DTM10 as an example.

⚠ CAUTION

- Make sure to properly connect the connection cord.
- If the PC to be connected does not have an RS-232C connector, use a device that converts the D-Sub9 pin to USB, etc.

Nitto Kohki has confirmed connectivity with the following product. (We do not provide support.)
BSUSRC0610BS made by BUFFALO

DTM10



4. How to Use

Calculate the output torque (converted value) from the numerical data (converted current value) sent by the monitor. You can calculate the output torque (converted value) by sending the current value of the electric screwdriver motor to the PC or PLC.

To convert the converted current value to the output torque, it is necessary to calculate the torque conversion factor first. You can calculate the output torque (converted value) from the torque conversion factor by using the following formula.

$$\text{Output torque (converted value)} = \text{Converted current value} \times \text{Torque conversion factor}$$

Here it is explained with DTM10 as an example.

⚠ CAUTION

- **The calculated torque value is the output torque. It is not the fastening torque.**
Using a torque wrench, etc., make sure the screw is fastened at the target fastening torque by the loosening torque method or refastening torque method.
- **After confirming that the screw is fastened at the target torque using a suitable method, calculate the torque conversion factor.**
- **Be sure to test more than once based on actual screw fastening conditions (screw, workpiece and other conditions).**
Perform the test periodically.
- **When the torque or rotational speed of a current control type electric screwdriver is changed, measure the conversion factor in each case.**
Conversion factor will be different for each setting. Measure the conversion factor several times.

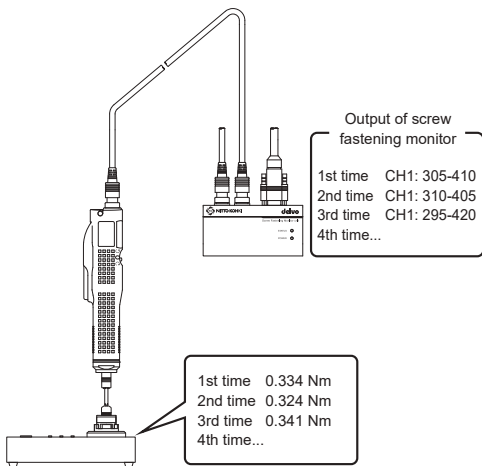
1 Fasten the screw with a torque checker (torque sensor) or workpiece.

Fasten the screw several times while making sure the screw is fastened at the target torque.

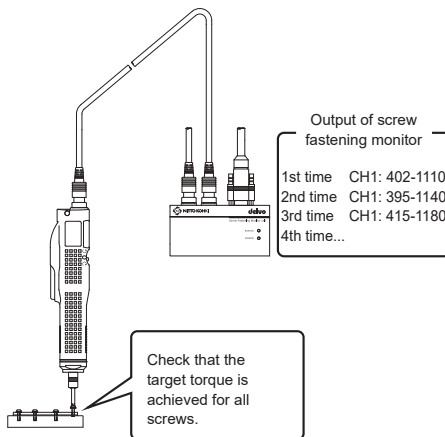
2 Record the converted current value.

If you are using a torque checker (torque sensor), record the fastening torque.

Torque checker (torque sensor)



Workpiece



3 Calculate the torque conversion factor.

$$\text{Torque conversion factor} = \frac{\text{Output torque average (target value)}}{\text{Average of converted current value of screw fastening monitor}}$$

E.g.: Calculation with a torque checker (torque sensor) (screw fastened 3 times)

$$\frac{\begin{array}{c} \text{1st time} \quad \text{2nd time} \quad \text{3rd time} \quad \text{Screw fastening frequency} \\ \downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ (0.334 + 0.324 + 0.341) \div 3 \end{array}}{\begin{array}{c} (305 + 310 + 295) \div 3 \\ \uparrow \\ \text{Torque conversion factor} \end{array}} = 0.0010978022$$

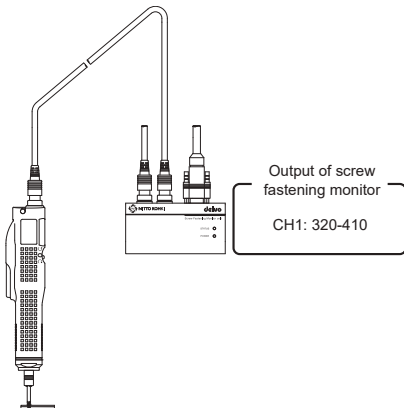
E.g.: Calculation with the target torque (target torque 1 Nm)

$$\frac{\begin{array}{c} \text{Target torque} \\ \downarrow \\ 1 \text{ Nm} \end{array}}{\begin{array}{c} (402 + 395 + 415) \div 3 \\ \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \\ \text{1st time} \quad \text{2nd time} \quad \text{3rd time} \quad \text{Screw fastening frequency} \end{array}} = 0.0024752475$$

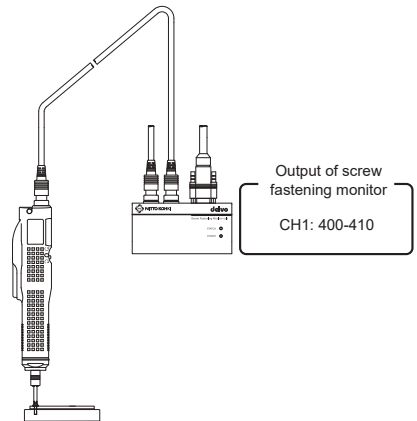
4 Based on the torque conversion factor, calculate the output torque from the converted current value at the time of actual screw fastening.

When calculating the conversion factor with the torque checker (torque sensor)

When calculating the conversion factor with the target torque



$$\begin{array}{l} 320 \\ \text{Converted} \\ \text{current value} \end{array} \times \begin{array}{l} 0.0010978022 \\ \text{Torque conversion} \\ \text{factor} \end{array} = \begin{array}{l} 0.351 \text{ Nm} \\ \text{Output torque} \end{array}$$



$$\begin{array}{l} 400 \\ \text{Converted} \\ \text{current value} \end{array} \times \begin{array}{l} 0.0024752475 \\ \text{Torque conversion} \\ \text{factor} \end{array} = \begin{array}{l} 0.99 \text{ Nm} \\ \text{Output torque} \end{array}$$

Using time and rotation signal

⚠ CAUTION

- **Rotation signal is output only by DTM45.**
- **DTM45 outputs time or rotation signal.**
You can select to output either of the two in "Measurement method" (MEASURE) of the common settings of the controller (DCC0241X-AZ). For controller settings, check the instruction manual of DCC0241X-AZ.

You can check whether the numerical values of time and rotation signal to be sent by the monitor are within the maximum/minimum values of screw fastening to determine that the screw is fastened correctly.

⚠ CAUTION

- **To fix the output numerical value, use the same operations method to perform the screw fastening.**
- **Periodically measure the time and rotation signal.**
- **The screw length or electric screwdriver speed may show individual differences. Also it changes due to the effects of heating or mechanical loss, etc.**
Measure several times.
- **Consider the screw length tolerance, workpiece individual differences, and operations scattering while setting the electric screwdriver operation.**
Examples of setting the time are given below. Do the same for the rotation signal. (For example -10%/+10%)
Screw fastening time lower limit value: -10% from minimum measured value
E.g.) Minimum measured value 0.6 s → Set the screw fastening time lower limit value to 0.54 s
Screw fastening time upper limit value: +10% from maximum measured value
E.g.) Maximum measured value 0.7 s → Set the screw fastening time upper limit value to 0.77 s
- **There is an error difference in time measurement of the controller and screw fastening monitor in DTM10.**

1 Perform screw fastening several times using the electric screwdriver and record the maximum and minimum values of time and rotation signal sent by the monitor.

The numerical data of time and rotation signal is sent by the monitor to a PC/PLC.

Data is not sent if stoppage occurs before torque up or at reverse rotation.

Calculate the upper and lower limit values of time and rotation signal from the minimum and maximum values.

2 Check on the PC or PLC whether the time and rotation signal sent by the monitor are within the maximum/minimum values of actual screw fastening.

In the following conditions, screw fastening is considered as not done appropriately.

Relation between the output data from the monitor and the actual values	Causes
Lower limit value of time/rotation signal > Actual value	Short screw, refastening, screw galling, etc.
Upper limit value of time/rotation signal < Actual value	Long screw fastened mistakenly, etc.

Software for confirmation of output torque

Nitto Kohki provides a software for PCs which can be used to confirm the output torque and save history. For details, see our website.

<http://www.nitto-kohki.co.jp/e/>

5. Appendix

When an abnormality has occurred (Troubleshooting)

Symptoms	Location to investigate	Solution
Unable to receive data	Is there a mistake, error, or a disconnection in the RS-232C cable?	Select an appropriate cable (cross or straight) and check the connection.
	Is the communication setting done appropriately?	Review the communication setting on the PC or PLC.
The electric screwdriver does not operate	Is the connection cord connected to the proper place?	Check the connection of the connection cord.

Maintenance and inspection

WARNING

- Always perform inspection before using the tool.
- Do not disassemble or alter the tool.
- Use genuine parts.

CAUTION

- **For repair or part replacement, contact your dealer where you purchased the tool.**
Repairing requires special knowledge and skills. If repair is performed at a place other than a specialty store, the tool may not demonstrate its full performance or it could lead to an accident or injury.
- **Request repair with the failed status kept intact.**
When requesting a repair, do not throw away damaged parts. It could be important information for investigating the cause of the failure so do not change the status.

Inspection locations	Caution
Cord	Failure to perform inspection could result in fire or electric shock. ● Check if the cord is damaged and if found stop using the tool. ● Do not store the cord by wrapping it around the main unit. If it has been stored with the cable wrapped around the main unit, immediately change the storage method.
Main unit	● Check for damage, cracks or breaks on the main unit.
Transmission data	● Check that the operation channel, converted current value and screw fastening time have not changed.
Maintenance	● If the main unit is stained, use a cloth soaked in soapy water and wrung out well to wipe off the stains. The tool does not have a waterproof structure and if water enters inside it could fail.

Disposal

- Separate power tools, accessories, and packing materials for environmentally-friendly recycling.
- Do not dispose of the power tool as household garbage.
- When disposing of electric tools, hand them over to Nitto Kohki or your dealer.



Separately-sold products

The following products are sold separately. To purchase these items, contact your dealer where you purchased your electric screwdriver.

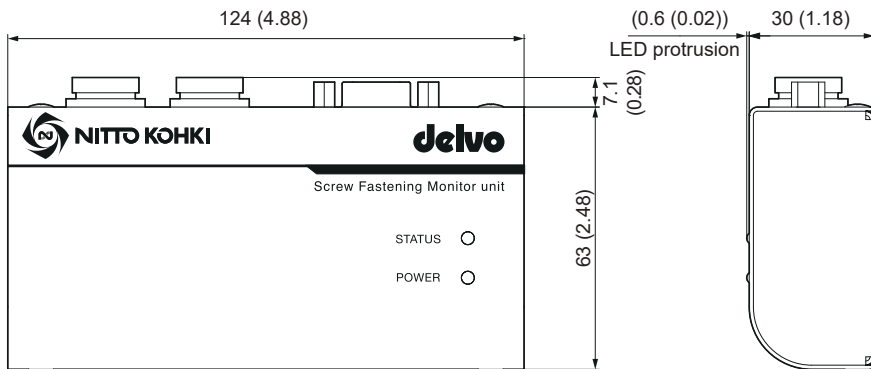
Product name (model)	Appearance	Specifications, etc.
Communication cable (straight) DLW9092		Used for connection to PC/PLC (p. 6)
Communication cable (cross) DLW9093		

Glossary

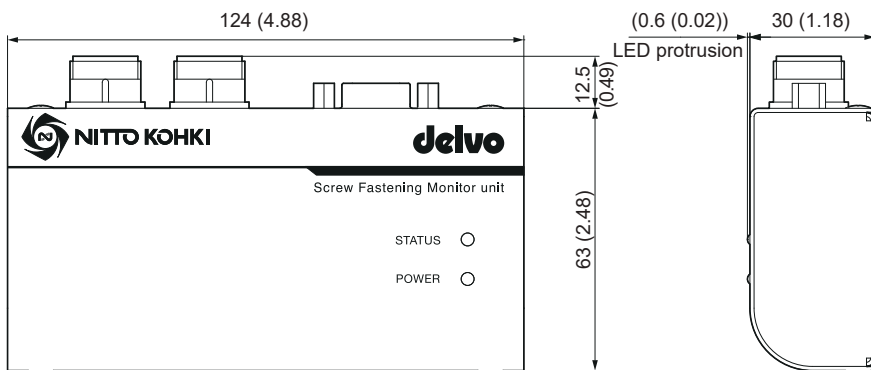
Term	Description
C	
Converted current value	Motor current value at torque up (the torque reaches the set torque and the electric screwdriver stops automatically) converted to a value between 0 to 4095
L	
Loosening torque method	Torque measurement method for measurement of torque value when the fastened screw is loosened with a torque wrench, and the screw starts rotating
O	
Operation channel	Channel during operations (while settings are valid)
P	
PLC	(Programmable Logic Controller) Electronic device performing sequential control according to a program
R	
Receptacle	Connector for connecting the connection cord
Refastening torque method	Torque measurement method for measurement of torque value when the fastened screw is further fastened with a torque wrench and the screw starts rotating again
T	
Torque up	The torque reaches the set torque and the screwdriver stops automatically
Torque conversion factor	Factor for calculating the output torque (converted value) from the converted current value
Torque checker	Torque measuring instrument

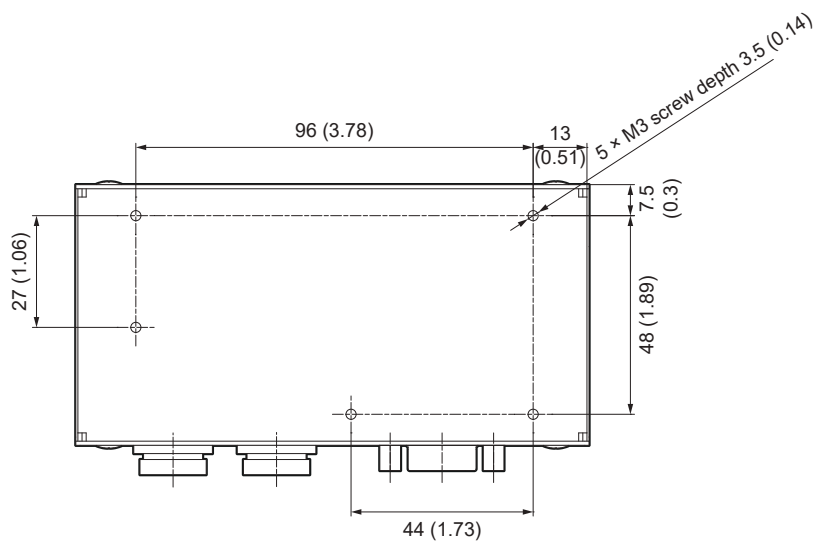
External dimensions

DTM10



DTM45



Back (common)



■ Overseas Affiliates / Offices

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